CSE36301 Machine Learning Assignment2

Deadline: Sunday, October 19, 2025, 11:59 PM

1 Probability Basics

In a tennis tournament, there are 80 players. Of these 80 players, 20 of them are at an advanced level, 40 of them are at an intermediate level, and 20 of them are at a beginner level. You randomly choose an opponent and play a game.

Your win rates against each level are as follows: 30% against advanced players, 60% against intermediate players, and 85% against beginner players.

1. What is the probability that you will play against an advanced player?
2. What is the probability that you will play against an intermediate player?
3. What is the probability that you will play against a beginner player?
4. You randomly choose an opponent and play a game. What is the probability of winning?
5. You randomly choose an opponent and play a game and you win. What is the probability that you won against an advanced player?

2 Random Variables

The probability mass function $p(x)$ of a discrete random variable X is given as

$$p(x=-2)=a, \quad p(x=0)=2a, \quad p(x=1)=2a, \quad p(x=4)=a. \tag{1}$$

1. Find the value of a.
2. Write the range (support) of the random variable X.
3. Draw the graph of $p(x)$.
4. Calculate the cumulative distribution function $F(x)$ of the random variable X, and draw it.
5. Calculate $E[X]$ and $\mathrm{Var}(X)$.
6. Let $Y = 2X + 1$. Find the PMF of Y and calculate $E[Y]$.
7. If you observe X twice independently (call them X_1 and X_2), what is $P(X_1 + X_2 = 2)$?

3 Viewing Ridge Regression as Augmented Least Squares

In this problem, we will show that ridge regression estimates can be obtained by ordinary least squares regression on an augmented data set.

3.1 Part A: Simple Example

Consider a simple case with $n = 3, p = 2$:

$$\mathbf{X} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} \tag{2}$$

1. Write down the normal equation for ordinary least squares.
2. Now augment the data with $\sqrt{\lambda}\mathbf{I}_2$ where $\lambda = 1$:

$$\mathbf{X}_{\text{aug}} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{y}_{\text{aug}} = \begin{pmatrix} 3 \\ 1 \\ 2 \\ 0 \\ 0 \end{pmatrix} \tag{3}$$

   Write down the normal equation for this augmented system.

3. Calculate $\mathbf{X}_{\text{aug}}^T\mathbf{X}_{\text{aug}}$ and $\mathbf{X}_{\text{aug}}^T\mathbf{y}_{\text{aug}}$ explicitly.

3.2 Part B: General Form

For a general centered matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ and $\lambda > 0$:

1. Define the augmented matrix $\mathbf{X}_{\text{aug}}$ and vector $\mathbf{y}_{\text{aug}}$ in terms of $\mathbf{X}$, $\mathbf{y}$, and λ.
2. Write the normal equation: $\mathbf{X}_{\text{aug}}^T\mathbf{X}_{\text{aug}}\boldsymbol{\beta} = \mathbf{X}_{\text{aug}}^T\mathbf{y}_{\text{aug}}$.
3. Expand the left-hand side: Show that $\mathbf{X}_{\text{aug}}^T\mathbf{X}_{\text{aug}} = \mathbf{X}^T\mathbf{X} + \lambda\mathbf{I}$.
4. Expand the right-hand side: Show that $\mathbf{X}_{\text{aug}}^T\mathbf{y}_{\text{aug}} = \mathbf{X}^T\mathbf{y}$.

3.3 Part C: Connection to Ridge Regression

1. Show that the normal equation from Part B gives:

$$(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y} \tag{4}$$

2. Explain why this is exactly the ridge regression solution.
3. Interpret the "artificial data points" $(\sqrt{\lambda}\mathbf{I}, \mathbf{0})$ in terms of model constraints or regularization.

3.4 Part D: Conceptual Understanding

1. Why do we set the response values of the artificial data points to zero?
2. How does this augmentation relate to the idea of "shrinking coefficients toward zero"?
3. What happens to the solution as $\lambda \to \infty$? Explain both algebraically and conceptually.

4 Logistic Regression vs. LDA

In this question, we will try to intuitively understand the advantages and disadvantages of logistic regression and LDA.

1. Describe the core assumptions behind logistic regression and LDA. Then state why logistic regression's assumption is often considered less restrictive.
2. If the Gaussian assumptions are correct, LDA is said to be more *statistically efficient* than logistic regression. Explain what the term *statistically efficient* means in this context.

3. Observations far from the decision boundary (which logistic regression effectively down-weights) still matter for LDA. Why is that, and what is one advantage and one disadvantage of this behavior?

4. Even unlabeled observations sometimes contain useful information. Name *which* parameters unlabeled data can inform in the LDA setting, and give a real-world scenario where unlabeled data are abundant but labels are expensive.

5. In a binary classification dataset, we say there is **perfect separation** when there exists at least one linear decision boundary that classifies all training examples with zero error. How can the LDA marginal likelihood act like a *regularizer* compared to logistic regression? Explain using the case of **perfect separation**.

6. In many real datasets, the Gaussian/equal-covariance assumptions are at least partially violated or some features are categorical. Which method would you choose as a default in such cases, and why?

5 Naive Bayes, KNN, and Evaluation Metrics

In this section, you will implement Naive Bayes, KNN classifiers, and evaluation metrics from scratch. Start your work from the attached `Assignment2.ipynb` file.

5.1 Naive Bayes

Task: In your `.ipynb` file, complete the TODO sections in the following methods in the `NaiveBayesFromScratch` class:

- `_gaussian_probability`
- `predict_proba`

Validation: Run the next two cells to train your model and validate your implementation. If your implementation is correct, the results should be similar to the scikit-learn implementation.

5.2 KNN

Task: In your `.ipynb` file, complete the TODO section in the following method in the `KNNFromScratch` class:

- `predict`

Validation: Run the next four cells to train your model with cross-validation and validate your implementation. If your implementation is correct, the results should be similar to the scikit-learn implementation.

5.3 Evaluation Metrics

Task: In your `.ipynb` file, complete the TODO section in the following method in the `ClassificationMetrics` class:

- `precision_recall_f1`

Validation: Run the next two cells to calculate metrics and validate your implementation. If your implementation is correct, the results should be similar to the scikit-learn implementation.

5.4 Performance Comparison and Analysis

By running the remaining cells, you will be able to compare the performance of both approaches. Based on your results, answer the analysis questions provided in the notebook.

6 Submission Requirements

Your submission must be a `.zip` file containing the following files with the exact naming convention:
`Assignment2_[YOUR_STUDENT_ID]_[COMPONENT].[EXTENSION]`

6.1 Required Files

- `Assignment2_[YOUR_STUDENT_ID]_Theoretical.[pdf/jpg/png]` (Sections 1–4)
- `Assignment2_[YOUR_STUDENT_ID]_Programming.ipynb` (Section 5 code)
- `Assignment2_[YOUR_STUDENT_ID]_Programming.pdf` (Section 5 output)
- `Assignment2_[YOUR_STUDENT_ID]_Analysis.pdf` (Section 5 analysis questions)

Example:

- `Assignment2_20241234_Theoretical.pdf`
- `Assignment2_20241234_Programming.ipynb`
- `Assignment2_20241234_Programming.pdf`
- `Assignment2_20241234_Analysis.pdf`

6.2 Theoretical Problems (Sections 1–4)

Any submission format is acceptable (handwritten, tablet, LaTeX, etc.). If you are handwriting your answers, while the TA will make every effort to read handwritten solutions, please try to write neatly to avoid situations where your work cannot be recognized.

6.3 Programming Assignment (Section 5)

Submit all:

1. Your completed `.ipynb` file with all TODO sections implemented
2. A `.pdf` export of your notebook showing all code and output
3. A `.pdf` file with your written answers to the analysis questions

7 Contact

If you have any questions regarding this assignment, please email the TAs:

- Seokhoon Jeong: `shjd0246@unist.ac.kr`
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